

MACHINE

LEARNING COURSE

FROM COLLEGE TO IT INDUSTRIES
CONCEPTS



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TSAR IT PRIVATE LIMITED

CONCEPTS

1. INTRODUCTION TO MACHINE LEARNING
2. PYTHON FOR MACHINE LEARNING
3. MATHEMATICS FOR MACHINE LEARNING
4. DATA PREPROCESSING & EDA
5. SUPERVISED LEARNING
6. UNSUPERVISED LEARNING
7. ENSEMBLE METHODS & ADVANCED ML
8. INTRODUCTION TO DEEP LEARNING
9. MODEL DEPLOYMENT & ML PIPELINES
10. REAL-WORLD PROJECTS
11. INTERVIEW PREPARATION



ABOUT OUR COMPANY

At **TSAR IT Private Limited**, we are a cutting-edge software company dedicated to providing innovative digital solutions to businesses and organizations. Located in Andhra Pradesh, we specialize in a range of services and software products tailored to meet the unique demands of our clients.

we like challenges to solve and give solutions to your business growth and you can monitor your business and employees work easily,

We offer services for a wide range of fields such as software development, web development, graphic development, cross platform development etc. For more information navigate to our services section.

“Explore The Services We Offer For You” we are experts in these services, we have strong experience and technical expert team, we have our own German Technology server, our team will complete and give you project in time.

we will provide our services like how clients feel comfortable, we can handle any type of project with low cost and we are expert in install and maintaining projects.

N RAKESH - CEO

Internship Courses:

In addition to our product offerings, **TSAR IT Private Limited** also provides Internship Courses for aspiring software developers, engineers, and tech enthusiasts. Our internship programs offer hands-on experience and training in cutting-edge technologies, including:

- **Java Development**
- **Spring Boot & React Full-Stack Development**
- **AI & Machine Learning**
- **Cloud Computing**
- **Software Testing & QA**

These internships are designed to bridge the gap between academic learning and real-world software development, preparing the next generation of IT professionals.

At **TSAR IT Private Limited**, we believe in the power of technology to drive success, innovation, and transformation. With a dedicated team of experts, we are committed to delivering high-quality, customized software solutions while fostering new talent through our internship programs.

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MACHINE LEARNING COURSE SYLLABUS

1. INTRODUCTION TO MACHINE LEARNING

- 1.1. What is Machine Learning?
- 1.2. ML vs. AI vs. Deep Learning
- 1.3. Types of ML: Supervised, Unsupervised, Reinforcement Learning
- 1.4. Real-world Applications (Healthcare, Finance, Retail, etc.)
- 1.5. Ethical Considerations in ML

2. PYTHON FOR MACHINE LEARNING

- 2.1. Python Basics (Syntax, Data Structures)
- 2.2. NumPy for Numerical Computing
- 2.3. Pandas for Data Manipulation
- 2.4. Matplotlib & Seaborn for Data Visualization
- 2.5. Scikit-learn Overview

3. MATHEMATICS FOR MACHINE LEARNING

- 3.1. Linear Algebra (Vectors, Matrices, Dot Product)
- 3.2. Probability & Statistics (Distributions, Bayes' Theorem)
- 3.3. Calculus (Gradients, Derivatives, Optimization)

4. DATA PREPROCESSING & EDA

- 4.1. Handling Missing Data
- 4.2. Feature Scaling (Normalization, Standardization)
- 4.3. Categorical Encoding (One-Hot, Label Encoding)
- 4.4. Exploratory Data Analysis (EDA) Techniques
- 4.5. Outlier Detection

5. SUPERVISED LEARNING

5.1. Regression:

- Linear Regression
- Polynomial Regression
- Ridge & Lasso Regression
- 5.2. Classification:
- Logistic Regression
- Decision Trees & Random Forest
- SVM (Support Vector Machines)
- k-Nearest Neighbors (k-NN)
- 5.3. Model Evaluation Metrics:
- MSE, RMSE (Regression)
- Accuracy, Precision, Recall, F1-Score (Classification)
- Confusion Matrix, ROC-AUC

6. UNSUPERVISED LEARNING

6.1. Clustering:

- k-Means
- Hierarchical Clustering
- DBSCAN
- 6.2. Dimensionality Reduction:
- PCA (Principal Component Analysis)
- t-SNE

7. ENSEMBLE METHODS & ADVANCED ML

- 7.1. Bagging (Random Forest)
- 7.2. Boosting (AdaBoost, XGBoost, Gradient Boosting)
- 7.3. Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV)
- 7.4. Bias-Variance Tradeoff

8. INTRODUCTION TO DEEP LEARNING

- 8.1. Neural Networks Basics
- 8.2. Activation Functions (ReLU, Sigmoid, Tanh)
- 8.3. Backpropagation & Gradient Descent
- 8.4. Introduction to TensorFlow/Keras

9. MODEL DEPLOYMENT & ML PIPELINES

- 9.1. Saving & Loading Models (Pickle, Joblib)
- 9.2. Flask/Django for ML API Deployment
- 9.3. Docker Basics for Containerization
- 9.4. Cloud Deployment (AWS SageMaker, GCP AI Platform)

10. REAL-WORLD PROJECTS

- 10.1. Predictive Maintenance
- 10.2. Customer Churn Prediction
- 10.3. Image Classification (CNN Basics)
- 10.4. Sentiment Analysis (NLP Basics)

11. INTERVIEW PREPARATION

- 11.1. ML Case Studies
- 11.2. Common Interview Questions
- 11.3. Mock Interviews

For Registration process:

Step 1:

Open your Browser & paste the link given below

Step 2:

Given Your Details

Step 3:

Click Register Button

REGISTRATION LINK :<https://internship.tsaritservices.com/registrationForm>

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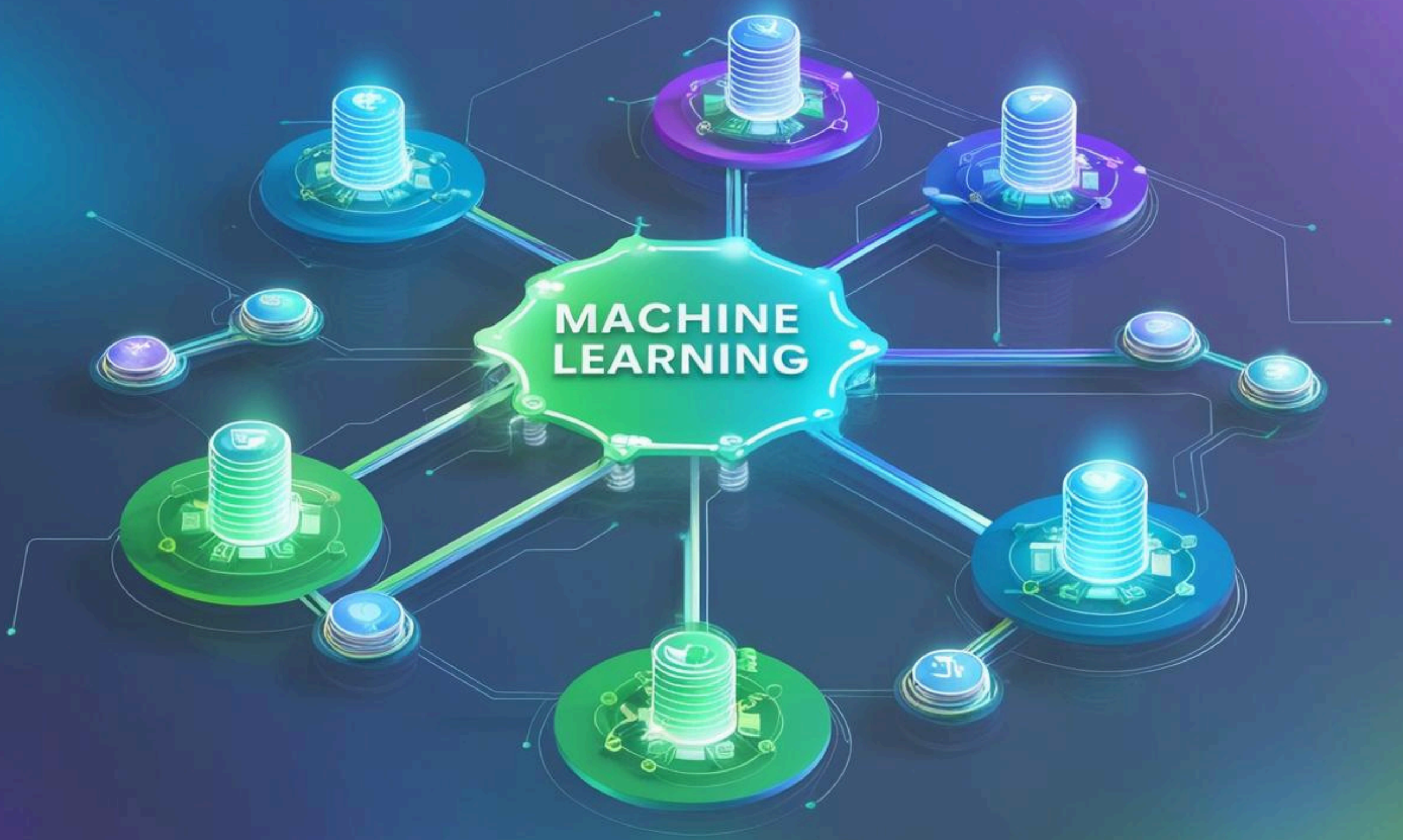
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